

OpenDSS and Typhoon HIL Co-Simulation for Real-Time Evaluation of a Distribution Network

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Abstract—Studies that require the analysis of large-scale distribution system models are usually performed using phasor based simulation packages. However, phasor based simulations can only perform steady-state analysis. With the advent of distributed generators (DGs), it is necessary to evaluate the impact of smart inverter controls on the distribution network in real-time. This paper evaluates the co-simulation platform between Typhoon Hardware-in-Loop (HIL) and OpenDSS by subjecting the inverter to various faults. First, the placement and sizing of DGs are determined using the Whale Optimization Algorithm (WOA). Next, the grid-tied photovoltaic (PV) inverter is simulated in Typhoon HIL and interfaced with the distribution network modeled in OpenDSS. The co-simulation platform is evaluated for its accuracy by comparing the results with those obtained from OpenDSS. Open Circuit faults are applied across the inverter terminals and their impact on the distribution network is analyzed in real-time. Additionally, the application of PV inverter control to improve the voltage profile of the distribution network is studied. Simulations carried out on the IEEE 13 bus unbalanced distribution network demonstrate the effectiveness of the co-simulation platform.

Index Terms—Distributed Generators (DGs), Smart Inverter, Whale Optimization Algorithm (WOA), Hardware-in-Loop (HIL)

I. INTRODUCTION

Global awareness of climate change has significantly fastened the penetration of renewable energy resources (RES). Several states in the U.S have established laws to mitigate greenhouse gas emissions. However, one of the major impediments to the large-scale deployment of photovoltaic and wind turbine systems is their lack of inertia. Grid-forming inverters (GFMI) offer a viable solution since they can operate autonomously and provide an emulated inertia. Since power semiconductor devices are prone to degradation, it is necessary to study the impact of their failure on the smart inverters and the microgrids to which they are connected. Moreover, the proliferation of RES in the distribution network necessitates the evaluation of smart inverter functionalities in real-time.

Existing phasor-based simulation packages such as OpenDSS, Gridlab-D, DlgSILENT, etc., lack real-time

evaluation capability. Though time-domain simulation packages such as PSCAD can be employed to study the impact of inverter controls on the distribution network, it would require high computational power and moreover real-time analysis is not possible. A co-simulation platform facilitates the real-time control of active distribution systems integrated with inverter-based resources (IBRs). OpenDSS and Typhoon is one such co-simulation platform based on the controller hardware-in-loop (CHIL) technology.

CHIL is a methodology to test the function of a system controller in real time. It is a low-cost method to test the power and control hardware external to the system. This is done by replacing the simulated control system with a real hardware controller which sends signals to the simulated power system. CHIL devices have several advantages such as low maintenance cost, unlimited expandability, automated testing and the high speed of controller development. These advantages have made CHIL devices very popular for testing the controller before developing the entire system. Several existing works in the literature discussed the application of HIL and CHIL platforms to test the power converter controllers and their impact on the power system models.

Diverse alternatives exist to implement a HIL platform. An overview of HIL and digital control techniques applicable to power converters for power factor correction is presented in [1]. Reference [2] presented the survey of state-of-the-art capabilities of the existing HIL, CHIL and power hardware-in-loop (PHIL) devices for testing power system protection and smart grid controllers. Moiz et al. [3] developed a power hardware-in-loop (PHIL) test bench consisting of a hardware under test and a low-voltage distribution grid modeled in real-time. The hardware under test can operate dynamically as a source or a load. PHIL capabilities using Opal-RT and an actual PV inverter are evaluated in reference [4] to analyze the voltage regulation of a distribution network. Authors in reference [5] developed a CHIL testbed to coordinate and control a group of distributed energy resources (DERs) for providing frequency

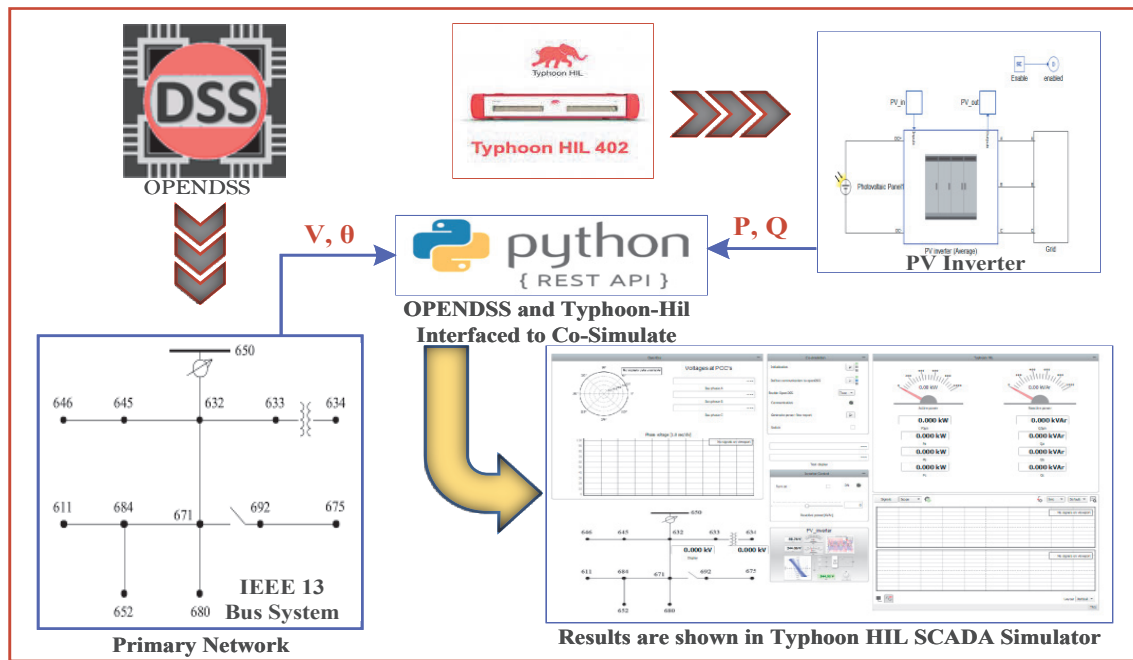


Fig. 1: OpenDSS Typhoon HIL Test Platform

regulation services. Bian et al. [6] developed a co-simulation platform that allows simultaneous simulation of communication and power system networks in real-time. The high variability of solar power requires dynamic simulations for PV impact studies using an electromagnetic transients (EMT) program. An open-source hybrid co-simulation using Python and OpenDSS is introduced in [7] to perform photovoltaic (PV) integration studies in the distribution systems. Electric vehicle (EV) smart charging is another area that requires real-time grid simulation in conjunction with EV charging hardware. A CHIL platform using Opal-RT and Typhoon HIL is implemented by Flack et al. [8] to validate the benefits of a decentralized charging algorithm on the voltage stability of a distribution system.

From the literature review, it is clear that significant progress has been made in developing diverse HIL platforms. However, though several HIL test benches exist currently, as discussed above, very few of them are based on open-source tools such as OpenDSS. In reference [9] authors utilized OpenDSS and Typhoon to design a HIL test bench to study the impact of PV inverter on the real-time reactive power control. However, the authors did not evaluate the results obtained from the co-simulation platform for accuracy. Moreover, the paper did not consider the impact of inverter failure on the distribution network. In this work, we discuss co-simulation applications using OpenDSS and Typhoon HIL. The primary goal of the co-simulation setup is to modernize the power engineering curriculum at the University of Missouri Kansas City (UMKC) and provide practical project experience to undergraduate students on the impact of inverter-based resources (IBRs) on the distribution network. In addition to studying the voltage

regulation of a distribution system using PV inverters, the significant contribution of this work is to analyze the impact of semiconductor switching failure on the distribution network.

The rest of the paper is organized as follows: section II presents the co-simulation architecture. The optimization algorithm to determine the optimal size and location of photovoltaic distributed generators (PVDGs) is discussed in section III. The simulation results obtained from the co-simulation platform are presented in section IV. Finally, the conclusions and future work are summarized in section V.

II. CO-SIMULATION ARCHITECTURE

The co-simulation setup between OpenDSS and Typhoon is shown in Fig. 1. The interface between OpenDSS and Typhoon is designed using Python APIs utilizing a voltage-power exchange over the power system bus. An unbalanced distribution network is modeled in OpenDSS whereas the three-phase PV inverter is emulated in the Typhoon HIL device. The power system bus for voltage-power exchange is a PQ bus on the OpenDSS side and it is emulated by a three-phase voltage source on the Typhoon HIL side. OpenDSS solves the snapshot power flow and updates the bus voltages after the PQ bus interface receives the active and reactive power injections from the PV inverter simulated in Typhoon. It then provides a voltage setpoint at the same bus interface to the PV inverter emulation running on the HIL device.

The optimum location and size of PVDGs are determined by formulating a multi-objective optimization algorithm in the fuzzy domain which is solved using WOA. The goal of the optimization problem is to improve the hosting capacity of PVDG, reduce the active power losses, and improve the

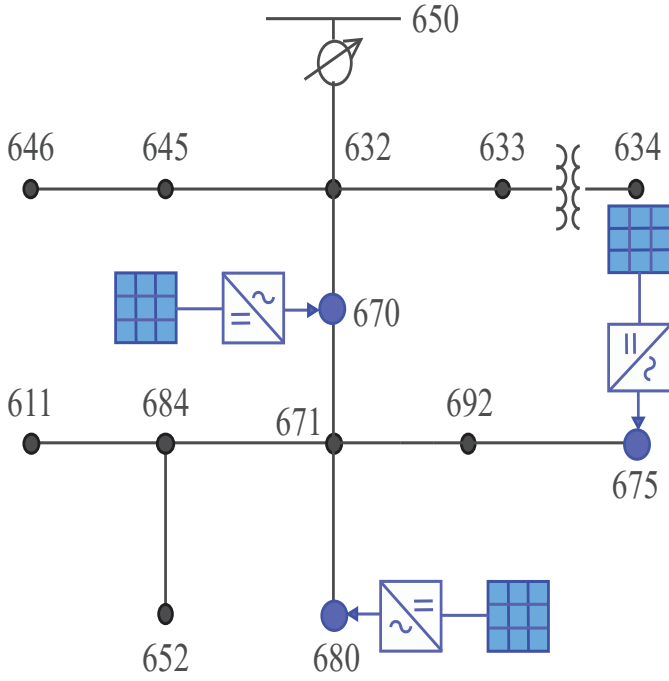


Fig. 2: IEEE 13-Bus Test Feeder with PVDGs

voltage profile of the distribution system as discussed in reference [10].

The proposed optimal planning model is validated using the modified IEEE 13-bus test feeder [11] through a co-simulation between Matlab and OpenDSS. Matlab 2022 is used to solve the WOA based optimization algorithm on an Intel Core i7-1185G7 with a 3.00 GHz CPU, 32 GB RAM, and a 64-bit windows operation system. The IEEE 13-bus test feeder is modeled in OpenDSS. The following table shows the location and size of the PVDGs and performance comparison with the base case is shown in Tab. II. Figure 2 shows the IEEE 13-bus test feeder with the three PVDGs.

TABLE I: Optimum Location and Size of PVDGs

Bus No	Size of the DG (kW)
680	649.9930
670	335.4716
675	647.5447

TABLE II: Performance Comparison

Performance Index	Base Case	With PVDGs
DG Penetration (kW)	-	1633.0093
Active Power Loss (kW)	112.3983	65.9649
Min Voltage, Va	0.9753	0.9965
Min Voltage, Vb	1	1
Min Voltage, Vc	0.9608	0.9805

III. SIMULATION RESULTS

To validate the co-simulation platform, Typhoon HIL, OpenDSS and MATLAB are used to obtain the simulation results. The 13-bus test system is modeled in OpenDSS and

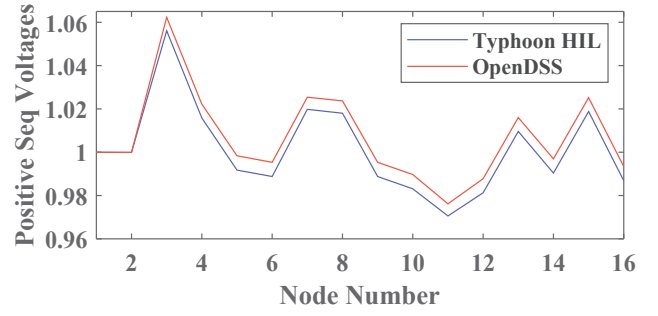


Fig. 3: Bus Voltage Comparison Between Typhoon HIL and OpenDSS

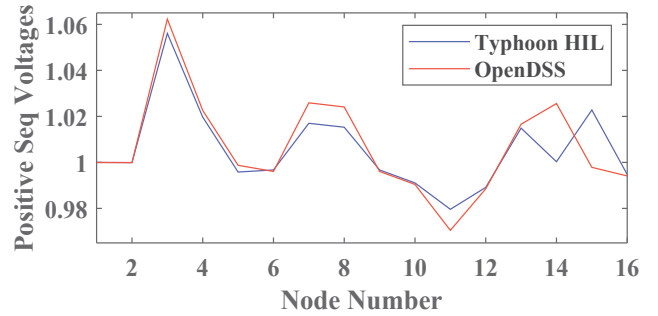


Fig. 4: Bus Voltage Comparison Between Typhoon HIL and OpenDSS With Reactive Power Injection

the PV inverters are modeled in Typhoon HIL. The inverter in Typhoon HIL is developed using the average model since it results in a faster simulation time. As mentioned in section II, the interface between Typhoon HIL and OpenDSS is developed using Python APIs. The simulation output data obtained from Typhoon HIL is exported into a mat file format and plotted using MATLAB.

Simulations are also carried out through OpenDSS-MATLAB co-simulation in which the PVDGs are modeled as generators. A comparison is made between the simulation results obtained from HIL-DSS and OpenDSS-MATLAB co-simulation platforms. Fig. 3 compares the co-simulation platforms for the positive sequence bus voltages in which the PV inverters in Typhoon HIL and DGs in OpenDSS are operating at the unity power factor. The comparison yields a maximum acceptable error of 0.66%. Fig. 4 compares the positive sequence bus voltages in which the PV inverters and DGs inject a reactive power of 302 kVAR, yielding a maximum error of 2.5%. The error in both the cases can be attributed to the oscillations observed at the maximum power point (MPPT) of the PVDGs modeled in Typhoon. It has to be noted that the DGs are modeled in OpenDSS without the MPPT algorithm which does not represent a practical scenario.

Figures 5 and 6 demonstrate the comparison between active and reactive power drawn by the loads. In Fig. 5, the PV inverters/PVDGs operate at unity power factor and in Fig. 6, the PV inverters/PVDGs inject a reactive power of 302 kVAR. A maximum error of 1.9 kW is observed when the

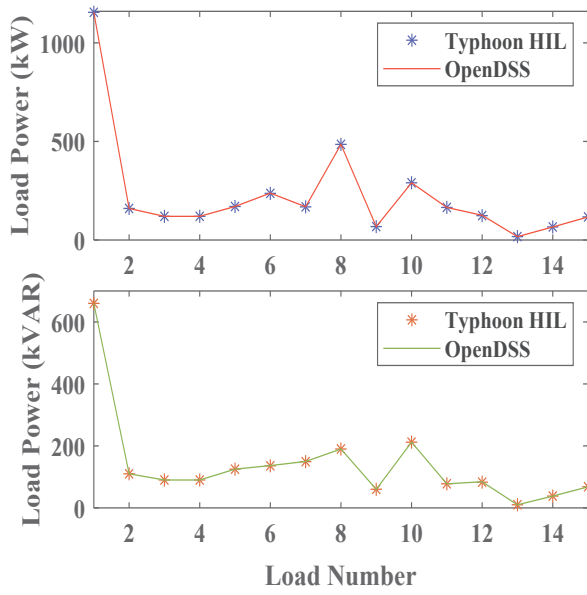


Fig. 5: Load Power Comparison Between Typhoon HIL and OpenDSS

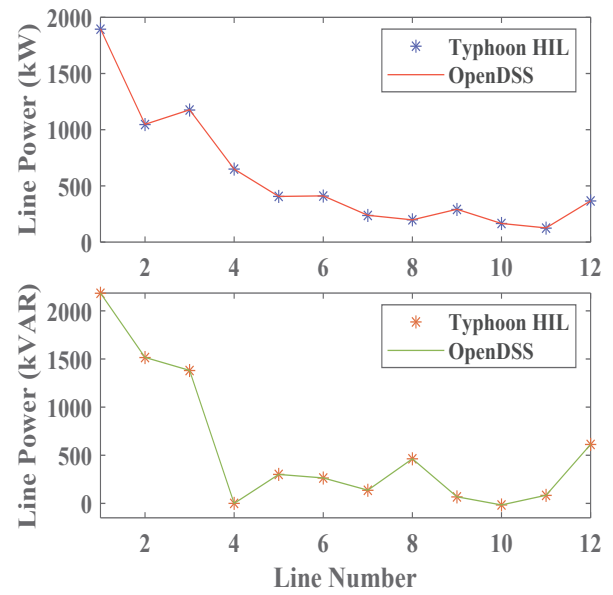


Fig. 7: Line Power Comparison Between Typhoon HIL and OpenDSS

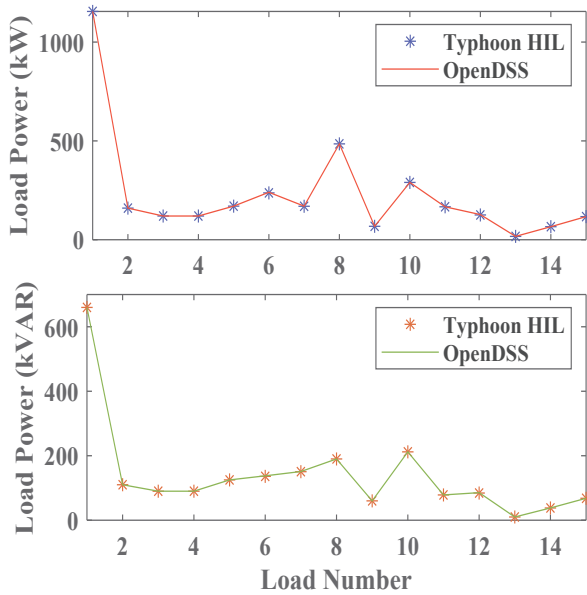


Fig. 6: Load Power Comparison Between Typhoon HIL and OpenDSS With Reactive Power Injection

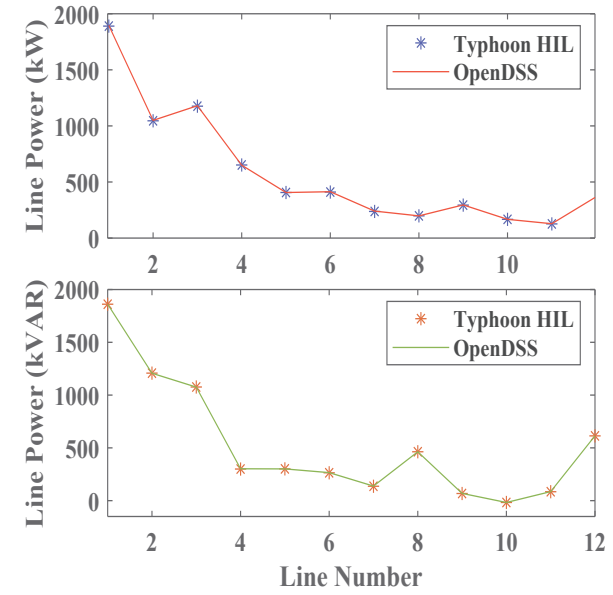


Fig. 8: Line Power Comparison Between Typhoon HIL and OpenDSS With Reactive Power Injection

PV inverters/PVDGs operate at unity power factor and the maximum error in the case of reactive power injection is 3 kW. Compared to the power drawn by the loads, the maximum error is within an acceptable range.

The comparison between the active and reactive power flowing through the lines is illustrated in Figures 7 and 8. In Fig. 7, the PV inverters/PVDGs operate at unity power factor and in Fig. 8, the PV inverters/PVDGs inject a reactive power of 302 kVAR. It is observed that the unity power factor operation yields a maximum error of 3 kW and PV inverters/PVDGs

operation with reactive power injection yields a maximum error of 6.7 kW. The error can be attributed to the active and reactive power injected by the PV inverters/PVDGs into the distribution lines. The results presented in figures 5 to 8 show that there is a certain degree of correlation between HIL-DSS and OpenDSS-MATLAB co-simulation platforms when the PV inverters/PVDGs are operating at unity power factor.

One of the advantages of the Typhoon HIL-OpenDSS co-simulation platform is to study the impact of PV inverter open circuit faults which cannot be implemented using the co-

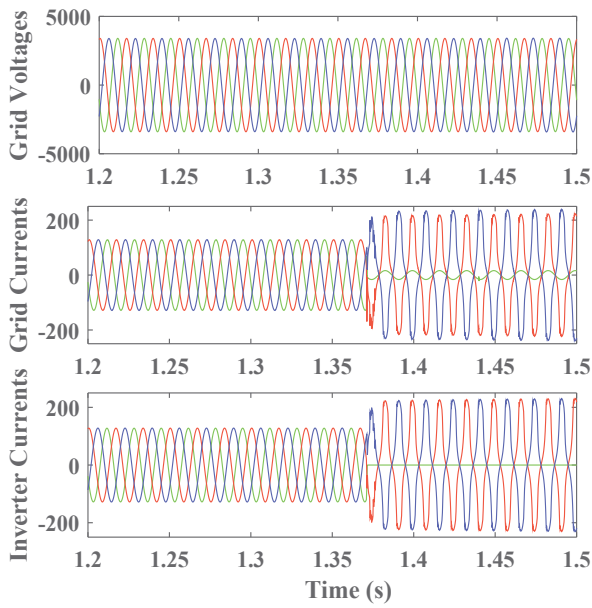


Fig. 9: Inverter Phase A Fault

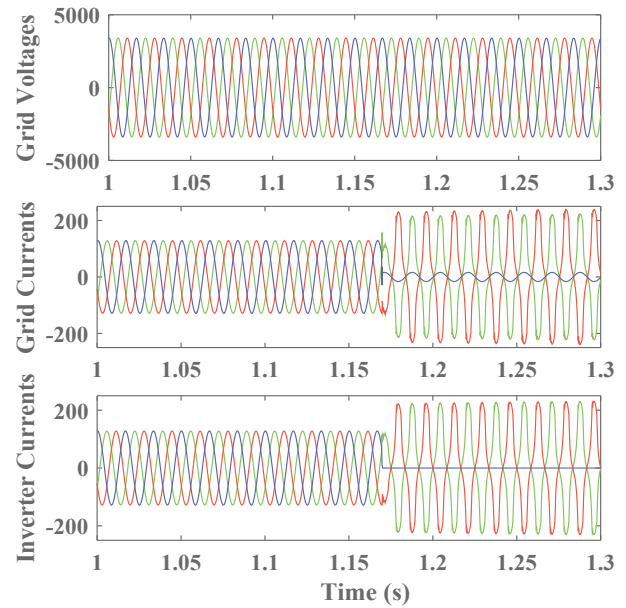


Fig. 11: Inverter Phase C Fault

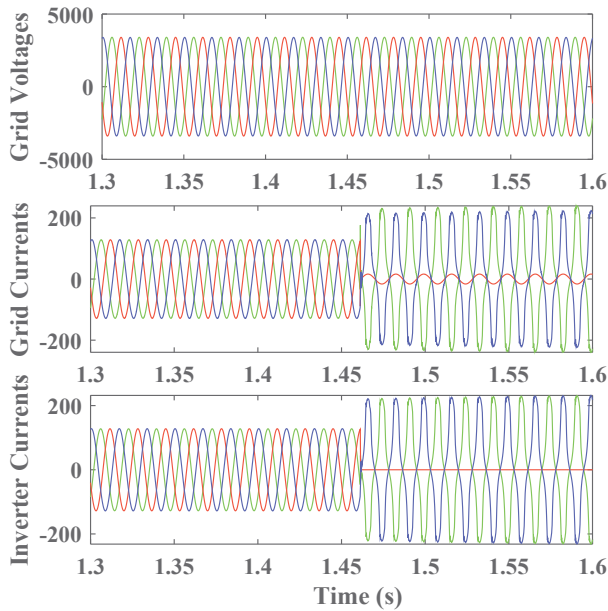


Fig. 10: Inverter Phase B Fault

simulation between OpenDSS and MATLAB. The PV inverter open circuit fault could be due to the failure of IGBTs as a result of aging. The failure of IGBTs is simulated by opening the switches in each phase and the results are demonstrated in figures 9 to 11. As shown in Fig. 9, the failure of IGBTs in phase A reduces the magnitude of the phase A grid current and the inverter current to zero when the fault is simulated at $t = 1.375$ s. It can also be observed that the currents in phases b and c increase after the occurrence of open circuit fault in phase a. Similar results can be observed in figures 10 and 11 when the open circuit fault is simulated in phases b and c.

Finally, the impact of PV inverter open circuit faults on the distribution bus voltages is demonstrated in Fig. 12. As shown in the figure, an open circuit fault reduces the PV inverter's output power, which leads to a drop in voltages at nodes 15 (bus 632) and 6 (bus 671). Similar results can be demonstrated for the remaining bus voltages in the IEEE 13-bus test system.

Fig. 13 presents a case study that depicts the impact of PV inverter control on the distribution bus voltages. In the base case scenario (without PV) of the IEEE 13 bus system, the bus voltages, as depicted in Fig. 13, comply with the NERC standards, with values ranging between 0.95 pu and 1.05 pu. However, when the loads are increased by 1.5 times, the voltage levels at $t=40$ sec decrease and fall outside the NERC limits. To rectify this, PV1 is activated at $t=60$ sec and PV2 at $t=65$ sec, bringing the bus voltages back within the acceptable range. In order to enhance stability further, PV3 is enabled at $t=70$ sec. Next, the load is augmented by 2 times at $t=80$ sec using the Load Multiplier command in Typhoon Scada panel, resulting in voltages remaining within the specified range. At $t=100$ sec, the load is further increased by 2.5 times using the load multiplier, causing the bus voltages to drop to 2.21 kV. To restore the voltages within the acceptable range, the reactive power for PV2 and PV3 is increased to 100 kvar and 200 kvar, respectively, at $t=125$ sec and $t=130$ sec. Finally, at $t=180$ sec, the load is returned to the base case, and the voltages at each bus remain within the acceptable limits, as illustrated in the figure.

The future objective of this research is to integrate battery energy storage (BES) into the distribution system to address voltage fluctuations caused by load variations and PV generation changes. The implementation platform developed in this study shows promise for deployment in various types of distribution system networks.

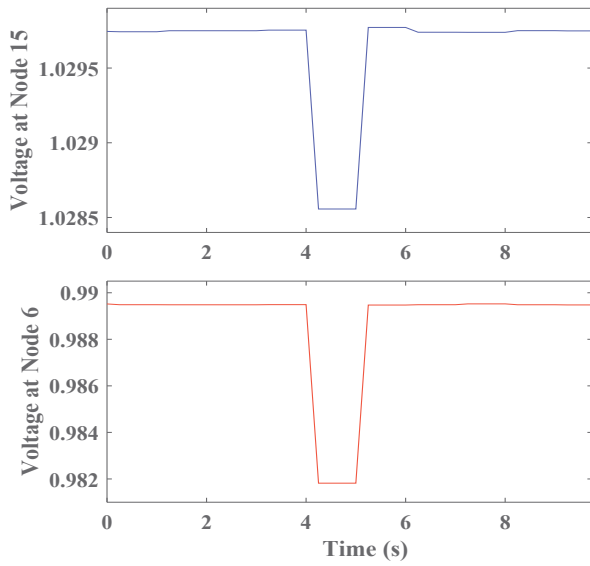


Fig. 12: Impact of Inverter Faults on the Bus Voltages

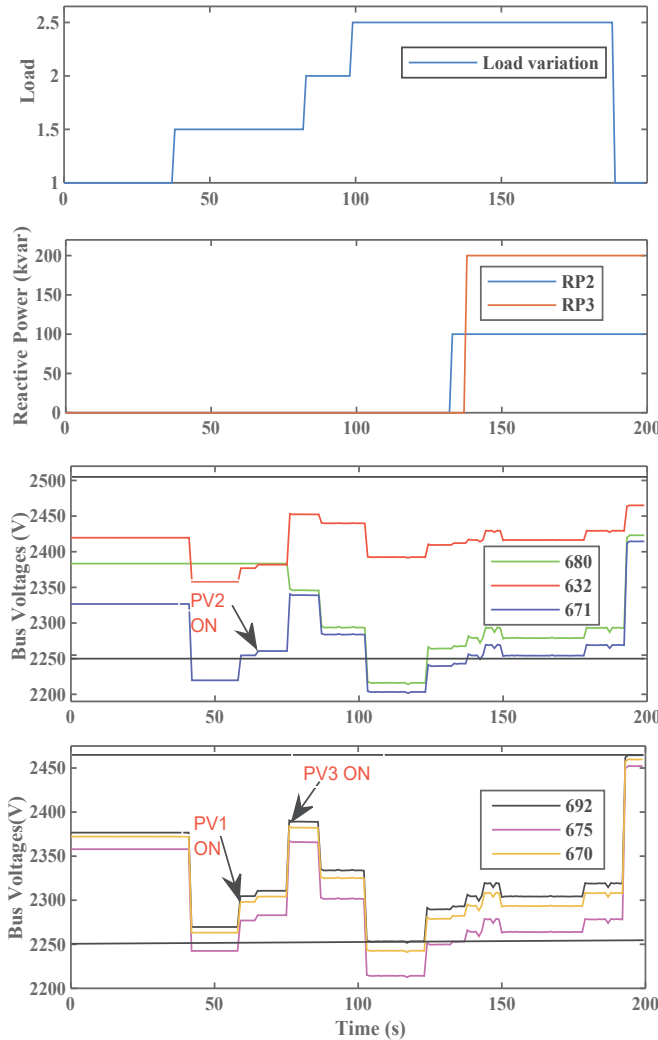


Fig. 13: Impact of PV Inverter Control on the Bus Voltages

IV. CONCLUSIONS AND FUTURE WORK

This paper demonstrated the functioning of the co-simulation simulation platform that has been developed using Typhoon HIL and OpenDSS. The results obtained from Typhoon HIL and OpenDSS co-simulation platform are compared with results obtained through a co-simulation between OpenDSS and MATLAB. The results demonstrate that the errors obtained for the bus voltages, load powers and line powers fall within an acceptable range with minimum deviations. The errors can be attributed to the MPPT algorithm and the average model of the inverter developed in Typhoon HIL. Future work consists of developing an accurate model of the inverter in Typhoon HIL and evaluating the co-simulation platform on much larger networks, such as the IEEE 123 and IEEE 8500 node test systems. In addition, we will study the impact of inverter switching on the power quality of the distribution network and investigate the applications of Vehicle-to-Grid (V2G).

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